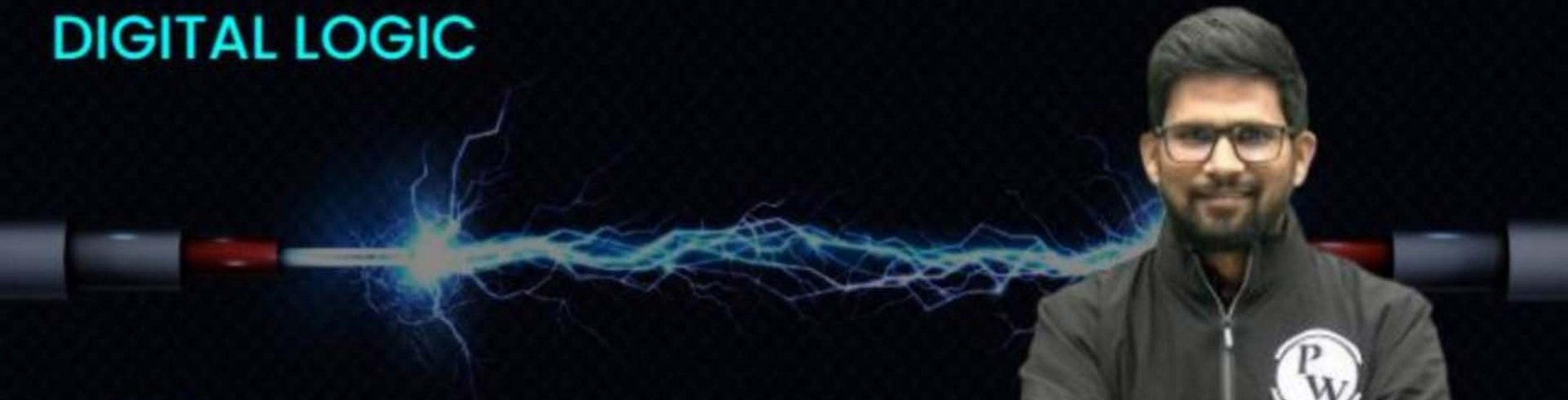
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 01

Miscellaneous Topics



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Recap of Previous Lecture



- Shift register
- Question Discussion



Numbers system

Q. Cosider a boolean function:

$f(W, x, y, z)$ such that:

$$f(W, x, y, z) = \sum(0, 1, 6, 7, 8, 9, 11, 13, 14, 15) + d\sum(2, 3)$$

$$\boxed{f(W, 0, 0, z) = 1} \longrightarrow f[0] = f[1] = f[8] = f[9] = 1$$

$$f(1, x, 1, z) = (x + z) \qquad f[10] = 0,\ f[11] = 1$$

$$f[14] = 1 = f(15)$$

$$f(W, 1, y, z) = Wz + y$$

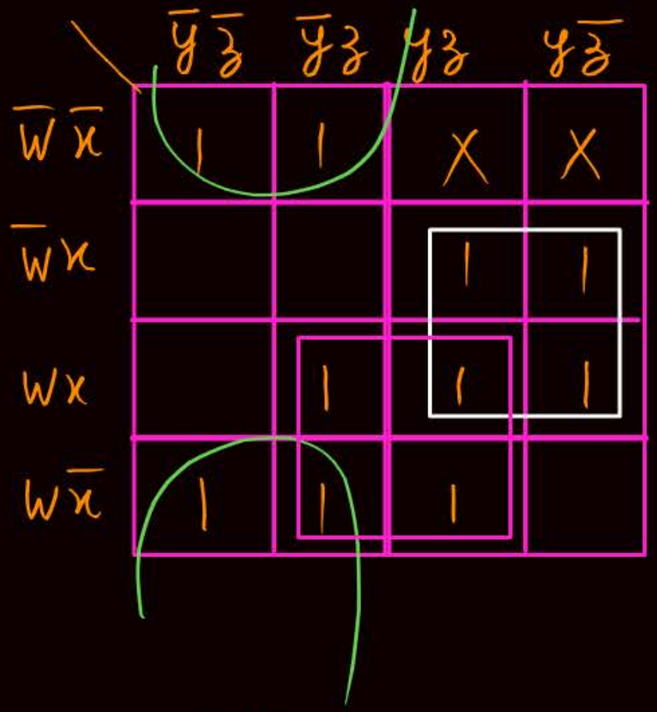
The no. of literals in minimal sum of of product expression of f is 6.

$f[4] = 0$ $\quad f[7] = 1$ $\quad f(15) = 1$

$f[5] = 0$ $\quad f[12] = 0$

$f[6] = 1$ $\quad f[13] = 1$

$f[14] = 1$



$$f(w,x,y,z) = \bar{x}\bar{y} + xy + wz$$

$$f(w,0,0,z) = 1 + 0 + wz = 1$$

Number System

Base of a number system : ⟶ base = r ⟶ $0 - (r-1)$ ⟶ digits can be used to write down any no. in base-r system.

decimal no system ⟶ base = 10 ⟶ (0 — 9) digits.

10^2	10^1	10^0
8	7	6

$= 8 \times 10^2 + 7 \times 10^1 + 6 \times 10^0$

$(10)_{10}$ ⟶ 10 is a number (2-digit number) in decimal no. system.

$(143)_5 = (5^2 \times 1 + 5^1 \times 4 + 5^0 \times 3)_{10}$ $\quad 5^2 5^1 5^0$

$= (48)_{10}$

base = 5 → (0 − 4) digit

$\times (162)_5$

$(377)_8 = [8^2 \times 3 + 8^1 \times 7 + 8^0 \times 7]_{10} = (255)_{10}$ $\quad 8^2 8^1 8^0$

$\times (876)_8$

→ 4 digit maximum no. in decimal no system $= (9999)_{10} = (10^4 - 1)_{10}$

4 ,, ,, ,, in Octal no. system $= (7777)_8 = [8^3 \times 7 + 8^2 \times 7 + 8^1 \times 7 + 8^0 \times 7]_{10}$

$= (8^4 - 1)_{10}$

$= (4095)_{10}$

→ 4 digit ,, ,, ,, $(0000)_8$ to $(7777)_8$

in base 5 $= (4444)_5$ $\quad 5^3 5^2 5^1 5^0$

$= (624)_{10} = (5^4 - 1)_{10}$

- Imp number system and their conversions :

Decimal no. system ⟶ base = 10 ⟶ (0 – 9) digits.

Binary no. system ⟶ base = 2 ⟶ (0 – 1) digits

Octal no. system ⟶ base = 8 ⟶ (0 – 7) digits

Hexadecimal system ⟶ base = 16 ⟶ (0 – 15) digits

10 — A	14 — E
11 ⟶ B	15 ⟶ F
12 ⟶ C	
13 ⟶ D	

$$\overset{8\,4\,2\,1}{(1\,1\,0\,1)}_2 = (8+4+0+1)_{10} = (13)_{10}$$

$$\overset{8^1\,8^0}{(3\,2)}_8 = (8\times 3 + 1\times 2)_{10} = (26)_{10}$$

$$\overset{16^1\,16^0}{(2A)}_{16} = (16^1\times 2 + 16^0\times 10) = (42)_{10}$$

$$\overset{16^1\,16^0}{(F\,F)}_{16} = [16^1\times 15 + 16^0\times 15] = [240+15] = (255)_{10} = (16^2-1)_{10}$$

$(25)_{10} = (011001)_2 = (31)_8$

$(337)_{10} = \begin{pmatrix} 8^2 & 8^1 & 8^0 \\ 5 & 2 & 1 \end{pmatrix}_8 = \begin{pmatrix} 16^2 & 16^1 & 16^0 \\ 1 & 5 & 1 \end{pmatrix}_{16}$

$(122)_{10} = 0(1111010)_2 = (172)_8$

$(77)_{10} = \begin{pmatrix} 5^2 & 5^1 & 5^0 \\ 3 & 0 & 2 \end{pmatrix}_5$

$(25)_{10} = \begin{pmatrix} 8^1 & 8^0 \\ 3 & 1 \end{pmatrix}_8$

$(122)_{10} = \begin{pmatrix} 8^2 & 8^1 & 8^0 \\ 1 & 7 & 2 \end{pmatrix}_8$

$(337)_{10} = (521)_8 = (101010001)_2$

$(25)_{10} = \begin{pmatrix} 16^1 & 16^0 \\ 1 & 9 \end{pmatrix}_{16}$

$(122)_{10} = (7\ A)_{16}$

Binary to Octal & Octal to Binary

$(111\,011\,010)_2 = (732)_8$

$(00\,010\,110)_2 = (126)_8$

$(00\,011\,.\,101\,100)_2 = (13.54)_8$

$(23.567)_8 = (010\,011\,.\,101\,110\,111)_2$

0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Binary to Hexadecimal & Hexadecimal to binary

$$0(0|0|0\,|0|1\,|1\,|0)_2 = (2AE)_{16}$$

$$(0|0|1|0|1\cdot|0|1|0|1|0)_2 = (5B\cdot B6)_{16}$$

$$(2AFB)_{16} = (0010\ 1010\ 1111\ 1011)_2$$

- $(236)_8 = (9E)_{16} = (0|0\,0|1\,|1\,0)_2$

$$(477)_{10} = \begin{pmatrix} 16^2 & 16^1 & 16^0 \\ 1 & D & D \end{pmatrix}_{16} = (000\ 1\ 1101\ 1101)_2$$

Question

A number $N_1 = (142)_5$ then it will be equal to $(xyz)_6$ then $(x + y + z)$ is ___(___7___)$_{10}$___. $(11)_6$

$N_1 = (142)_5 = (25 + 10 + 2)_{10} = (47)_{10} = (\overset{6^2}{1}\ \overset{6^1}{1}\ \overset{6^0}{5})_6$

$x = 1$

$y = 1$

$z = 5$

$$\begin{array}{r} 1 \\ +\ 1 \\ +\ 5 \\ \hline (11)_6 \end{array}$$

Question

A number $N_1 = (247.56)_8$ then its hexadecimal equivalent number is

(a) A7.B7 $= (010\ 100\ 111 \cdot 101\ 100)_2$

(b) A7.B8 $= (A7 \cdot B8)_{16}$

(c) (47.B8)

(d) B7.B8

Question

A binary no. is given as B = 00101010.11010 then its hexadecimal and octal equivalent will be :

(a) $(A2.D)_{16}, (52.64)_8$

(b) $(2A.D)_{16}, (52.64)_8$

(c) $(2A.D)_{16}, (25.64)_8$

(d) $(A2.D)_{16}, (25.64)_8$

Question

Two no. N_1 and N_2 are given as

$N_1 = (111)_2$, $N_2 = (777)_8$

Then their hexadecimal equivalent will be :

(a) $(7)_{16}$, $(1FF)_{16}$

(b) $(7)_{16}$, $(777)_{16}$

(c) $(F)_{16}$, $(777)_{16}$

(d) $(F)_{16}$, $(1FF)_{16}$

$= (111)_2$

$= (7)_{16}$

$= (001\ 111\ 111\ 111)_2$

$(1\ F\ F)_{16}$

Question

A relation is given as :

$(124)_x = (64)_y$

$x^2 x^1 x^0$ $\quad$ $y^1 y^0$

$x \geq 5$ & $y \geq 7$

Then minimum value of (x + y) _____14_____.

$(x^2 + 2x + 4)_{10} = (6y + 4)_{10}$

$x = 5 \rightarrow (39)_{10}$

$x = 6 \rightarrow (52)_{10}$

$(x)_{min} = 6$

$(y)_{min} = 8$

$y = 7$, $(46)_{10}$

$y = 8$ $(52)_{10}$

EVA

(6, 8) (10, 20), (12, 28)

(16, 48) (18, 60), (22, 88)

(24, 104) (28, 140) (30, 192)

$\Rightarrow y = \frac{x^2 + 2x}{6} = \frac{x(x+2)}{6}$

$x = 6 \rightarrow y = 8$

$x = 10 \rightarrow y = 20$

$x = 12 \rightarrow y = 28$ — infinite Soln.

$x = 16 \rightarrow y = 48$

$x = 18 \rightarrow y = 60$



Topic : 2 Min Summary

Number System

Thank you

GW

Soldiers !

